

E-COMMERCE Data Engineering

End-to-End Data Engineering & Business Intelligence Project



SQL

ETL

Data Warehouse

POWER BI

Swipe to see how it was built →

01 —Project Roadmap



This project demonstrates an End-to-End data engineering solution covering business analysis, data modeling, ETL, data warehousing, SQL analytics, and Power BI.



Business Requirement Analysis

Analyzed the business case, gathered requirements, and identified core business entities.



Database Design

Designed the ERD, resolved many-to-many relationships, and built a normalized (3NF) operational database..



Data Warehousing

Designed the dimensional data warehouse, implemented the Star Schema, and developed ETL pipelines across the Medallion layers.



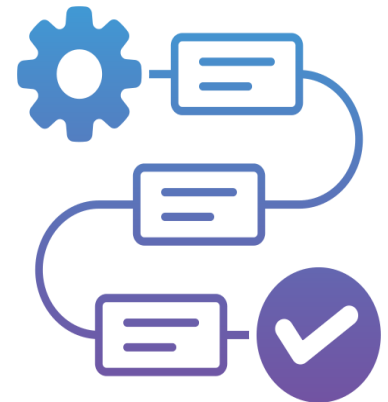
Analytics & BI

Developed analytical SQL queries, generated KPIs, and built an interactive Power BI dashboard.

02 — Project Workflow



- 1 Analyze Business Requirements
- 2 Extract Core Business Entities
- 3 Design Conceptual Data Model (ERD)
- 4 Resolve Many-to-Many Relationships Using Bridge Tables
- 5 Build Normalized 3NF Operational Database
- 6 Design Dimensional Data Warehouse
- 7 Implement Star Schema and SCD Type 2
- 8 Automate ETL Pipeline With SQL Server Agent
- 9 Build SQL Analytics & Power BI Dashboard



03 —Business Requirements Analysis



Objective

Identify the core business entities, relationships, and rules that drive the operational database and data warehouse.

Business Domains

Core Business

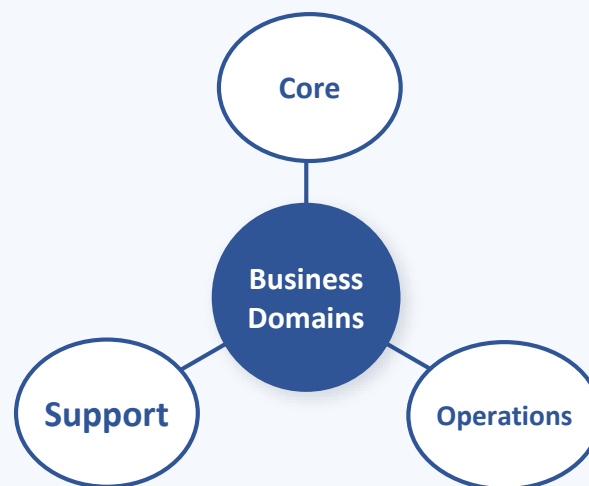
- Customer Management
- Product Management
- Order Processing

Operations

- Payments
- Shipping
- Inventory

Support

- Promotions
- Customer Reviews
- Employees & Departments
- Suppliers



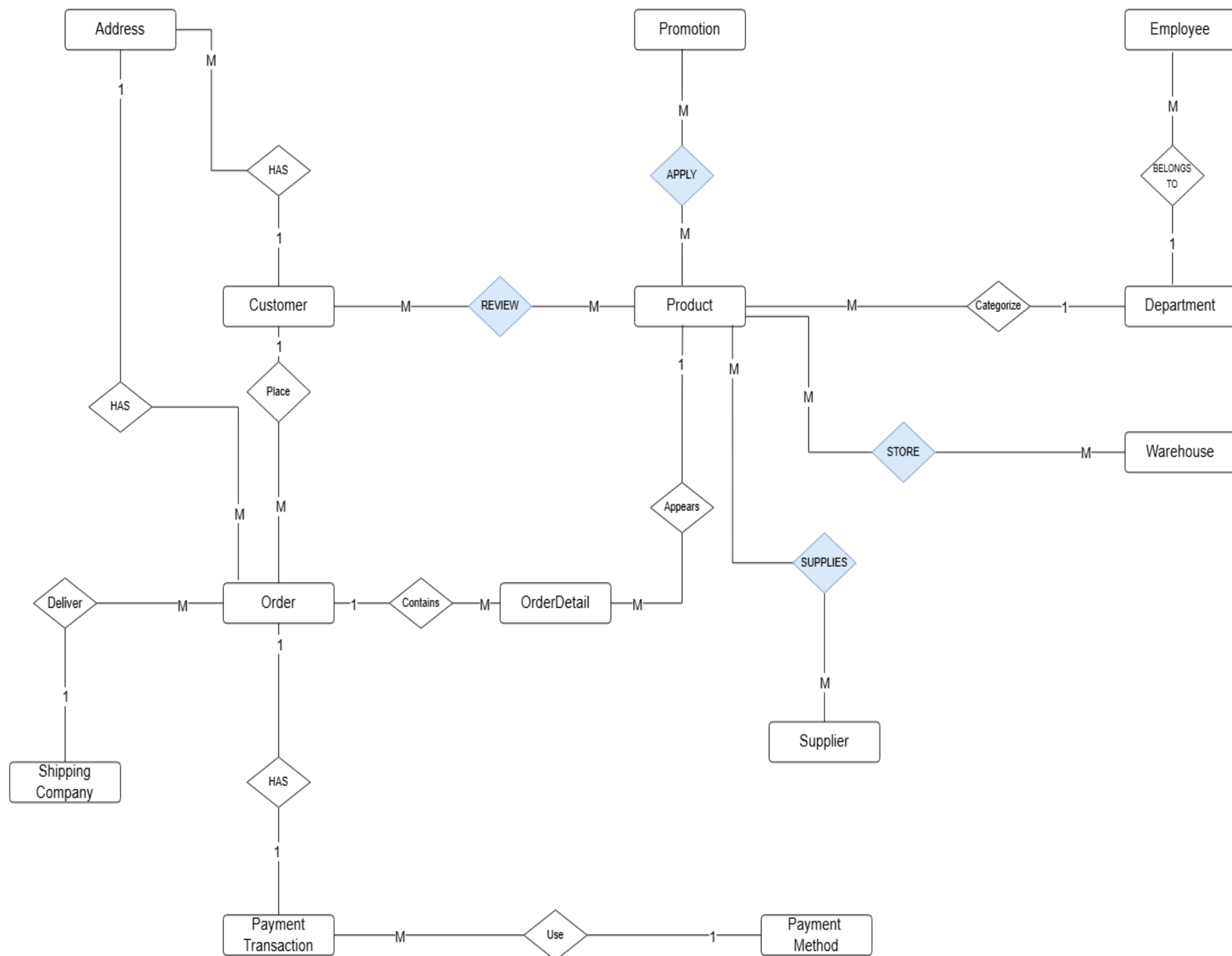
Business Analysis Outcomes

- ✓ Identified core business entities
- ✓ Defined business rules and workflows
- ✓ Identified entity relationships (1:1, 1:M, M:M)
- ✓ Resolved many-to-many relationships using bridge tables

04 —Conceptual Data Model (ERD)

Transforming business requirements into a structured conceptual data model.

[View the complete ERD on GitHub →](#)



Design Outcome

Translated business requirements into a conceptual ERD, defining the entities, relationships, and business rules required for the operational database.

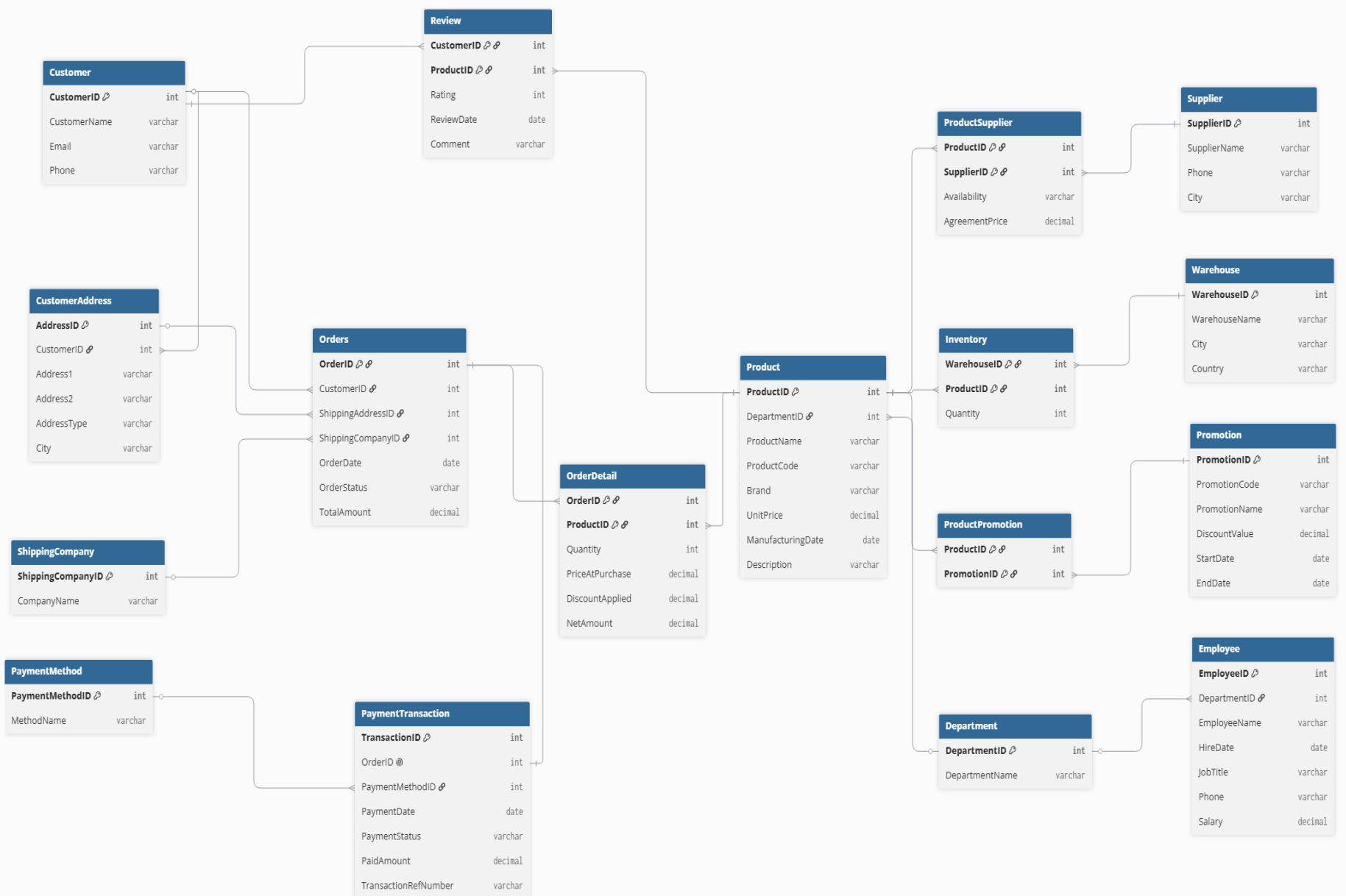
05 —Operational Database Design (3NF)

Transforming the conceptual ERD into a normalized operational database.



OLTP DATA MODEL

Normalized Transactional & Operational Database Schema



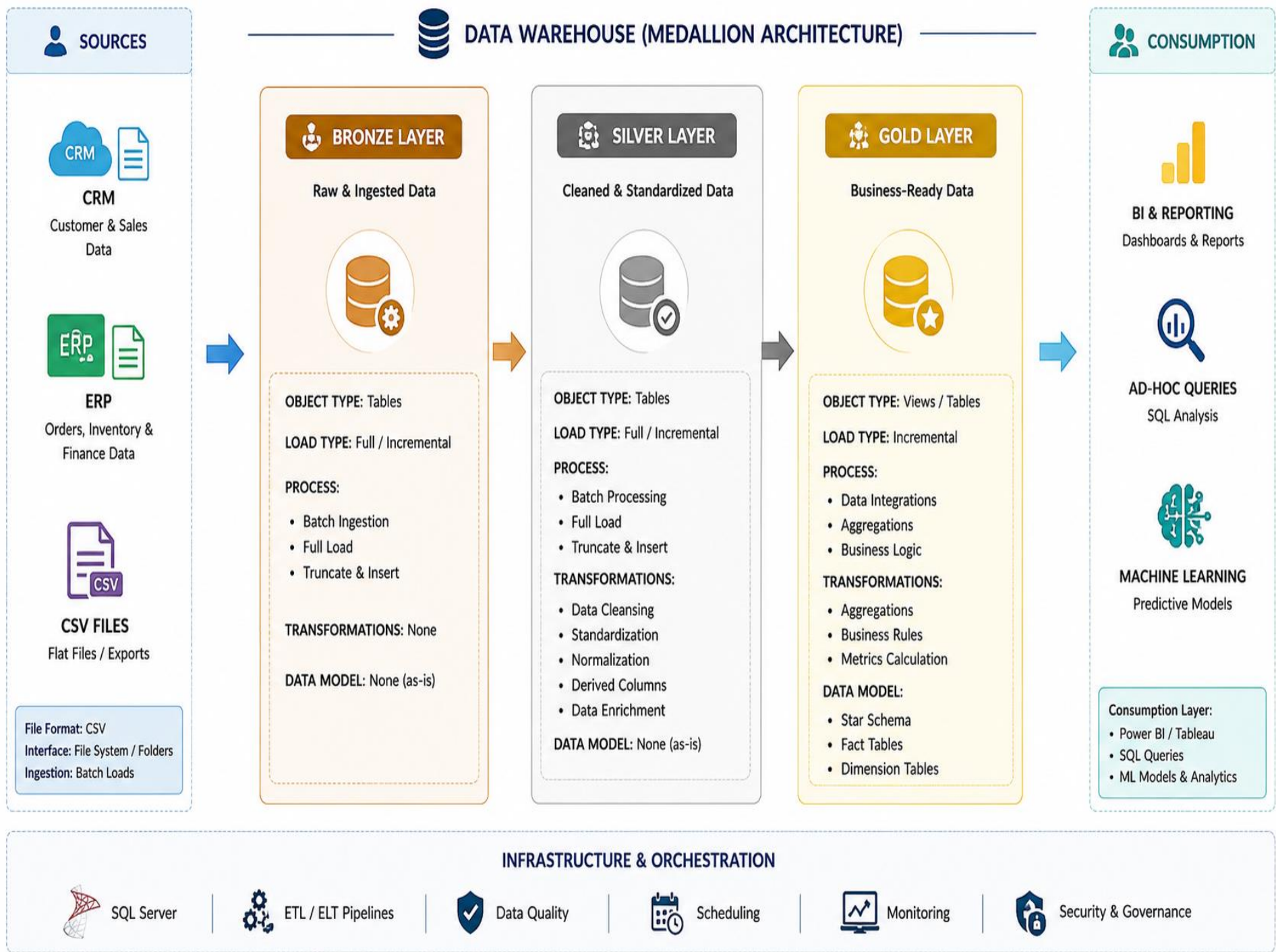
Design Outcome

Designed a normalized 3NF operational database that ensures data integrity and supports efficient transactional (OLTP) processing.

06 —Data Warehouse Architecture

Transforming transactional data into an analytics-ready data warehouse.

Designed using Medallion Architecture (Bronze, Silver, Gold)



Architecture Overview

A data warehouse was designed to integrate, transform, and prepare data for business intelligence, the Gold layer serves as the source for the dimensional Star Schema.

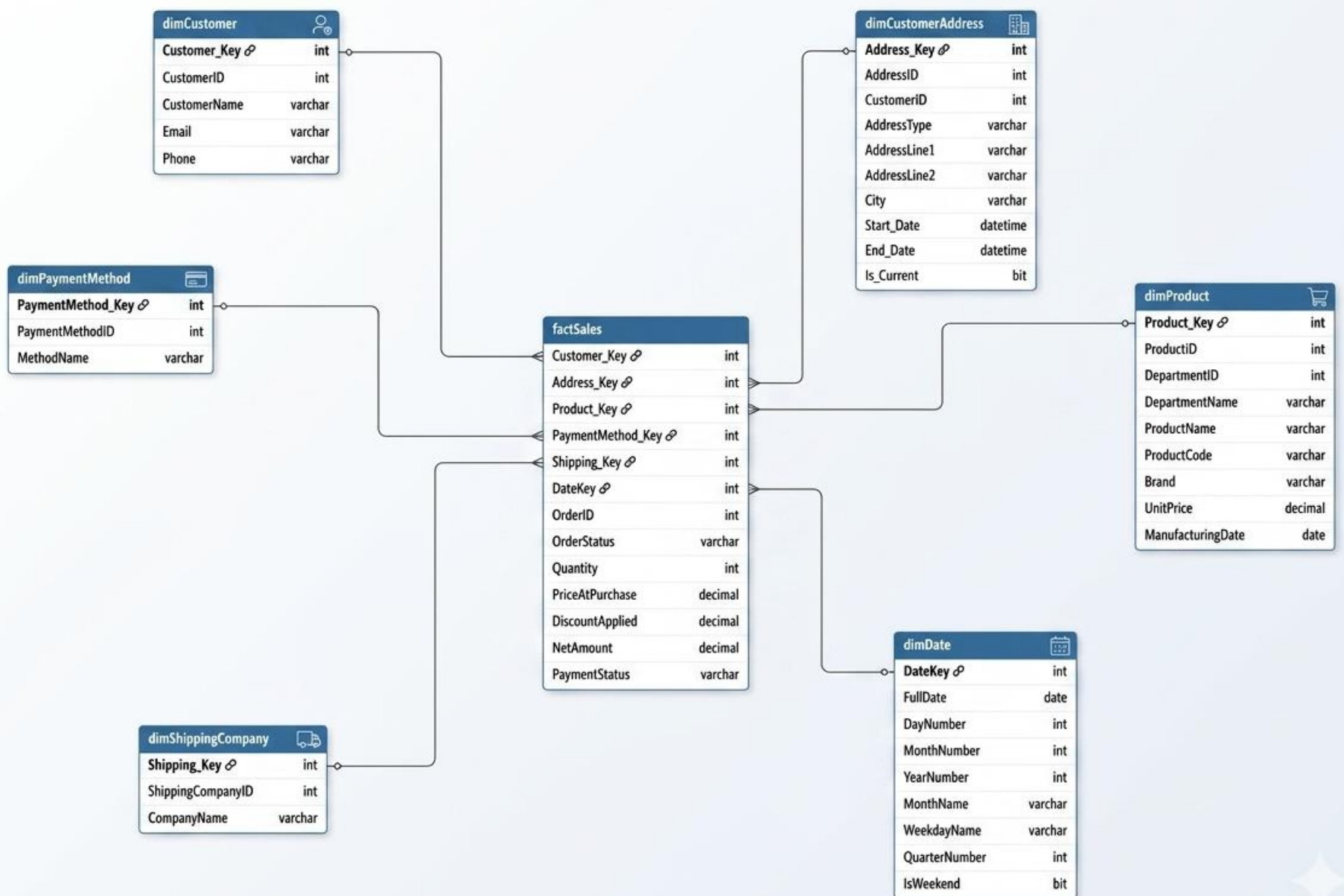
07 —Dimensional Data Model

Designing an analytics-ready model (Star Schema) for reporting and business insights.

Fact grain : One record for each product included in an order.



SALES DATA MART



Design outcome

The Gold layer was modeled as a Star Schema, creating an analytics-ready data model optimized for SQL analytics and Power BI reporting.

08 —Implementing SCD Type 2



Preserving Customer Address History with SCD Type 2

Instead of overwriting existing records, SCD Type 2 creates a new version for each change, preserving history for accurate reporting and analysis.

Business Scenario: A customer changes their shipping address from **Jazan** to **Abha**.

1 Current Customer Address

	AddressID	CustomerID	AddressType	AddressLine1	AddressLine2	City	dwh_create_date
1	11	5	Home	6788 Al Nakheel St.	Unit 72, Block A, Medina, Saudi Arabia	MEDINA	2026-07-17 13:55:59.3366667
2	12	5	home	2360 Al Olaya St.	Villa 67, Al Zahra District, Jazan, Saudi Arabia	JAZAN	2026-07-17 13:55:59.3366667

- Only the current address exists before the update.



2 Address Updated — History Preserved

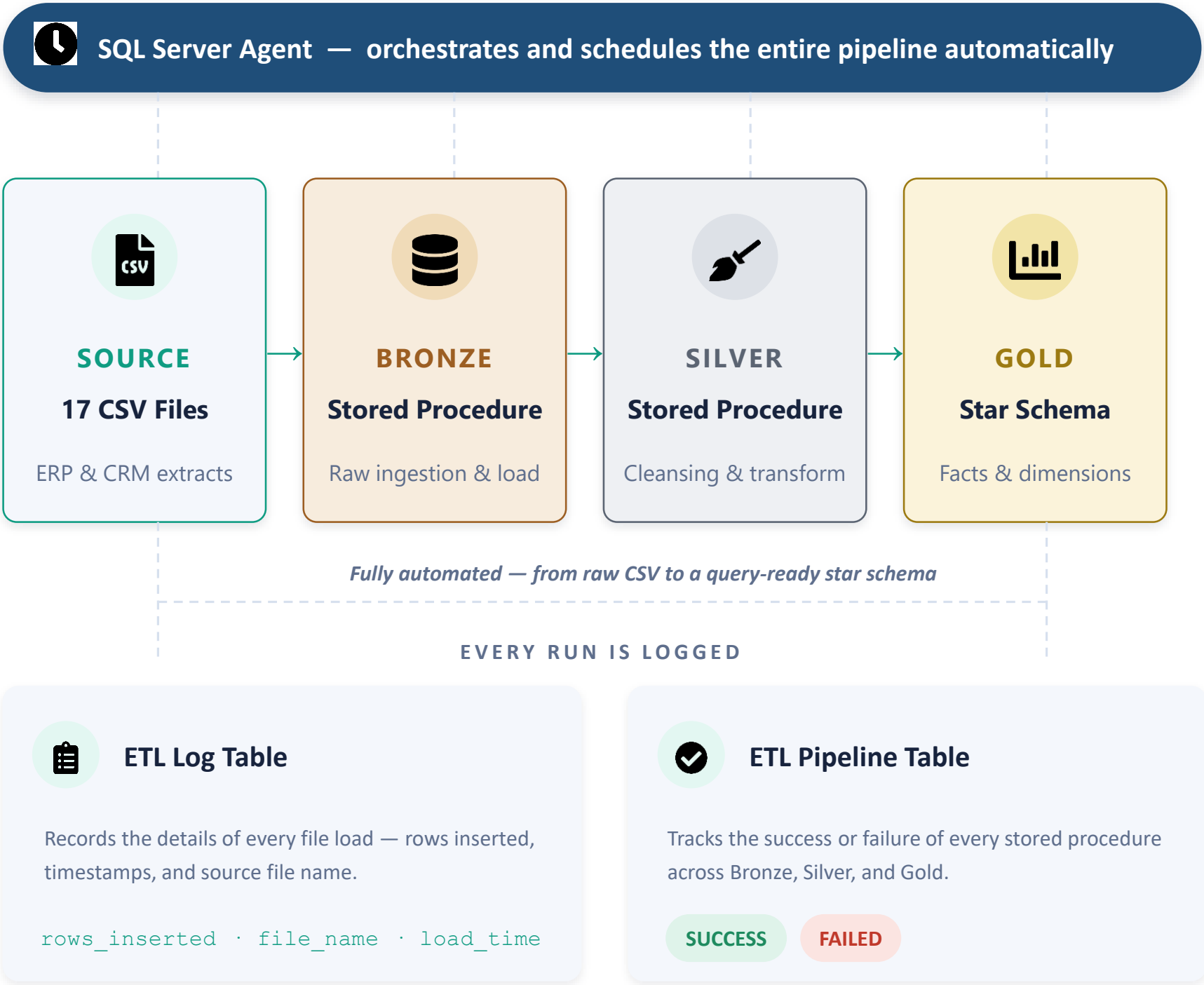
	Address_Key	AddressID	CustomerID	AddressType	AddressLine1	AddressLine2	City	START_DATE	END_DATE	Is_Current
1	11	11	5	Home	6788 Al Nakheel St.	Unit 72, Block A, Medina, Saudi Arabia	MEDINA	2020-01-01 00:00:00.0000000	NULL	1
2	12	12	5	home	2360 Al Olaya St.	Villa 67, Al Zahra District, Jazan, Saudi Arabia	JAZAN	2020-01-01 00:00:00.0000000	2026-07-17 13:57:22.4133333	0
3	10000	12	5	home	2360 Al Olaya St.	Villa 67, Al Zahra District, Jazan, Saudi Arabia	ABHA	2026-07-17 13:57:22.4233333	NULL	1

- A new version is inserted for the updated address .
- The previous record is closed by assigning an End Date and setting Is_Current = 0, while the new record becomes the active version.

09- ETL Pipeline



17 source files, moved and transformed through Bronze → Silver → Gold, with every run logged automatically.



Implementation Summary

Developed an automated ETL pipeline that ingests, transforms, and loads analytics-ready datasets for SQL analytics and BI reporting.

10— Inside the ETL Pipeline

17 source files, moved and transformed through Bronze → Silver → Gold, with every run logged automatically.

Pipeline Execution Log

Tracks every ETL procedure execution across Bronze, Silver, and Gold, including runtime, status, and error details.

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SELECT *

FROM etl.pipeline_log;

log_id	run_id	pipeline_name	step_name	start_time	end_time	status	error_message	
1	1	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Sales_Dw_ETL	bronze.load_bronze	2026-06-24 11:46:11.6884949	2026-06-24 11:46:13.0043419	Success	NULL
2	2	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Sales_Dw_ETL	silver.load_silver	2026-06-24 11:46:13.0043419	2026-06-24 11:46:14.6759436	Success	NULL
3	3	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Sales_Dw_ETL	gold.load_gold	2026-06-24 11:46:14.6759436	2026-06-24 11:46:16.1523265	Success	NULL
4	4	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Sales_Dw_ETL	gold.load_dim_CustomerAddress_SCD2	2026-06-24 11:46:16.1523265	2026-06-24 11:46:16.2650955	Success	NULL
5	5	2AEDD47B-A3D1-4AD3-982F-DEC9EE988255	Sales_Dw_ETL	bronze.load_bronze	2026-06-24 11:49:43.1676717	2026-06-24 11:49:43.9812043	Success	NULL
6	6	2AEDD47B-A3D1-4AD3-982F-DEC9EE988255	Sales_Dw_ETL	silver.load_silver	2026-06-24 11:49:43.9812043	2026-06-24 11:49:45.5173287	Success	NULL
7	7	2AEDD47B-A3D1-4AD3-982F-DEC9EE988255	Sales_Dw_ETL	gold.load_gold	2026-06-24 11:49:45.5173287	2026-06-24 11:49:46.7792821	Success	NULL
8	8	2AEDD47B-A3D1-4AD3-982F-DEC9EE988255	Sales_Dw_ETL	gold.load_dim_CustomerAddress_SCD2	2026-06-24 11:49:46.7795481	2026-06-24 11:49:46.8353966	Success	NULL
9	9	B2F7F760-809E-437E-887B-1AF00BCDB15D	Sales_Dw_ETL	bronze.load_bronze	2026-06-24 14:01:25.0695981	2026-06-24 14:01:25.7502542	Failed	Cannot bulk load because the file "D:\SC

File Load Log

Logs every source file load with inserted rows, timestamps, and execution status for complete pipeline traceability.

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SELECT *

FROM etl.file_load_log;

file_log_id	run_id	file_name	target_table	start_time	end_time	rows_loaded	status	error_message	
1	1	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Customer.csv	bronze.Customer	2026-06-24 11:46:12.1086859	2026-06-24 11:46:12.1207634	5020	Success	NULL
2	2	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	CustomerAddress.csv	bronze.CustomerAddress	2026-06-24 11:46:12.1207634	2026-06-24 11:46:12.1424162	9995	Success	NULL
3	3	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Department.csv	bronze.Department	2026-06-24 11:46:12.1424162	2026-06-24 11:46:12.1447447	5	Success	NULL
4	4	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Product.csv	bronze.Product	2026-06-24 11:46:12.1447447	2026-06-24 11:46:12.1517252	2000	Success	NULL
5	5	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Supplier.csv	bronze.Supplier	2026-06-24 11:46:12.1527260	2026-06-24 11:46:12.1650472	250	Success	NULL
6	6	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	ProductSupplier.csv	bronze.ProductSupplier	2026-06-24 11:46:12.1654008	2026-06-24 11:46:12.1758968	5944	Success	NULL
7	7	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Orders.csv	bronze.Orders	2026-06-24 11:46:12.1761497	2026-06-24 11:46:12.2872770	50000	Success	NULL
8	8	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	OrderDetail.csv	bronze.OrderDetail	2026-06-24 11:46:12.2872770	2026-06-24 11:46:12.8069519	149497	Success	NULL
9	9	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	PaymentMethod.csv	bronze.PaymentMethod	2026-06-24 11:46:12.8072190	2026-06-24 11:46:12.8097039	4	Success	NULL
10	10	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	PaymentTransaction.csv	bronze.PaymentTransaction	2026-06-24 11:46:12.8097039	2026-06-24 11:46:12.9119440	50000	Success	NULL
11	11	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	ShippingCompany.csv	bronze.ShippingCompany	2026-06-24 11:46:12.9129435	2026-06-24 11:46:12.9149434	6	Success	NULL
12	12	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Promotion.csv	bronze.Promotion	2026-06-24 11:46:12.9149434	2026-06-24 11:46:12.9169408	30	Success	NULL
13	13	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	ProductPromotion.csv	bronze.ProductPromotion	2026-06-24 11:46:12.9169408	2026-06-24 11:46:12.9209586	3047	Success	NULL
14	14	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Review.csv	bronze.Review	2026-06-24 11:46:12.9219385	2026-06-24 11:46:12.9689123	30000	Success	NULL
15	15	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Employee.csv	bronze.Employee	2026-06-24 11:46:12.9689123	2026-06-24 11:46:12.9719928	300	Success	NULL
16	16	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Warehouse.csv	bronze.Warehouse	2026-06-24 11:46:12.9722196	2026-06-24 11:46:12.9741847	25	Success	NULL
17	17	2B56ED9D-BBF3-446A-B9CC-CBC0C9165AF2	Inventory.csv	bronze.Inventory	2026-06-24 11:46:12.9741847	2026-06-24 11:46:13.0040899	7054	Success	NULL

11- SQL ANALYTICS



Analytical SQL queries built on the Gold layer using completed (PAID) transactions to answer key business questions.

1 BUSINESS KPI SUMMARY

```
93  -- Business KPI Summary
94
95  select * from gold.vw_business_KPIs
96
```

	measure_name	measure_value
1	Revenue	35003511.54
2	Total_Quantity	224258.00
3	Avg_Price	156.09
4	Average_Order_Value	2104.59
5	Total_Orders	50000.00
6	Total_Products	2000.00
7	Total_Customers	5020.00

2 TOP 5 PRODUCTS BY REVENUE

```
20  -- Which 5 products generate the highest revenue?
21
22  select top 5 p.ProductName , sum(f.NetAmount) as Total_Revenue
23  from gold.fact_sales f
24  left join gold.dim_product p
25  on f.Product_Key = p.Product_Key
26  where f.PaymentStatus = 'PAID'
27  group by ProductName
28  order by Total_Revenue desc
29
```

	ProductName	Total_Revenue
1	Philips Shaver Series	188091.15
2	Apple iPhone	182392.83
3	Philips Monitor	180135.03
4	Apple iPad Pro	179855.58
5	Apple MacBook Pro	178489.11

3 SALES PERFORMANCE OVER TIME

```
19  -- Analyze Sale Performance Over Time
20
21  --Yearly Revenue Analysis
22  select
23      d.Year , count(distinct Customer_Key) as Total_Customers , sum(f.NetAmount) as Total_Year_Revenue
24  from gold.fact_sales f
25  left join gold.dim_date d
26  on f.DateKey = d.DateKey
27  where d.DateKey <> -1 and f.PaymentStatus = 'PAID'
28
29  group by d.Year
30  order by Total_Year_Revenue desc
31
```

	Year	Total_Customers	Total_Year_Revenue
1	2024	2671	8137572.79
2	2022	2681	7935642.55
3	2021	2673	7853292.85
4	2023	2622	7762970.54
5	2025	685	1771535.97
6	2020	735	1542496.84

12- Power Bi Dashboard

An interactive report connected to the Gold layer — customer, product, and revenue insights in one view.



13- Project Summary:



Designed and implemented an end-to-end e-commerce data platform, starting with a normalized operational database and sample transactional data, then building a Data Warehouse with automated ETL, monitoring, SCD Type 2, SQL analytics, and an interactive Power BI dashboard.



35M+

SAR in Sales revenue tracked



50K

orders Processed



17

source files automated

Key Learnings:

- ✓ Translating business requirements into a scalable data model
- ✓ Built Operational Database (3NF)
- ✓ Designed a Medallion Data Warehouse
- ✓ Designed a Star Schema and implemented SCD Type 2
- ✓ Automated ETL Pipeline and Monitoring
- ✓ SQL Analytics and Power BI Dashboard



Explore the Full Project

Everything behind this project — the code, dashboard, and documentation — is ready for you to explore.



GitHub Repository

github.com/khaliedessam/End-to-End-E-Commerce-Data-Modeling-and-Analytics-Design



SQL Scripts

Queries, views, and schema



Documentation

Setup guide and project notes



Power BI Dashboard

Interactive report and visuals



Let's Connect on LinkedIn

linkedin.com/in/khalied-essam

Feedback and suggestions are always welcome.